BRANCHING & THE HEAD CURSOR

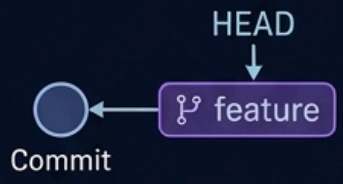
GIT BRANCHES

"A branch is a lightweight movable pointer to a commit."



WHAT IS HEAD?

"**HEAD** identifies your current position in Git, usually through the branch you have checked out."



MODERN COMMANDS

```
git switch feature
```

```
git restore app.js
```

Use `git switch` to change branches and `git restore` to discard or restore file changes. `git checkout` is the older multi-purpose command.

THREE MERGE OUTCOMES

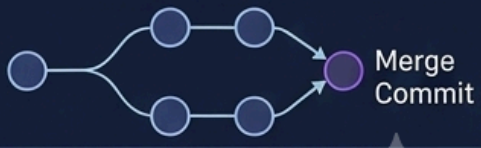
Branch Divergence
Two branches develop independently.



Fast-Forward Merge
The branch pointer moves forward when histories allow it.



Three-Way Merge
Git combines changes using a common ancestor.



A branch reference is a lightweight pointer, typically stored as a short text file containing a commit ID in loose-ref storage; packed refs may be stored differently.

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REMOTE SYNC & MERGE CONFLICTS

FETCH VS PULL

`git fetch`

Downloads remote updates and updates remote-tracking references. Does not merge them into your current branch.



`git pull`

Fetches remote updates, then integrates them into your current branch, usually by merging or rebasing depending on configuration.

CONFLICT ANATOMY

```
1 <<<<<<< HEAD (Current change)
2 const apiURL = "https://v1.api.com";
3 =======
4 const apiURL = "https://v2.api.com";
5 >>>>>>> feature-branch (Incoming change)
```

Git uses these markers to show conflicting sections. `HEAD` identifies the current side; the other side comes from the incoming change.

CLEAN FIX

1. Edit the conflicting file and choose the correct content.
2. Remove all conflict markers.
3. Run `git add filename`.
4. Complete the merge with `git commit`, if Git requires a merge commit.

💡 Test the result before committing.

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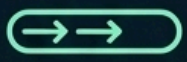
THE ULTIMATE GIT CHEAT SHEET

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SNAPSHOT MODEL

Git tracks project snapshots.



THREE STATES

Working tree -> Staging -> Repository.



BRANCH POINTERS

Branches point to commits.



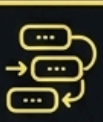
SWITCH & RESTORE

Switch branches; restore file changes.



MERGING TYPES

Fast-forward and three-way merge.



REBASE VS MERGE

Rebase replays commits; merge integrates histories.



REMOTE SYNC

fetch downloads updates; pull fetches and integrates.



STASHING WORK

git stash temporarily shelves changes.



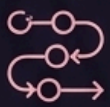
UNDOING MISTAKES

git revert creates undo commit; git reset moves a reference.



CONFLICT MARKERS

<<<<<<< ====== >>>>>>>



REFLOG SAFETY NET

git reflog helps find previous HEAD positions.



PRODUCTIVITY ALIASES

Create shortcuts with git config alias.

Save this guide for your next coding session.



TOP GIT INTERVIEW QUESTIONS

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QUESTION

WHAT IS THE DIFFERENCE BETWEEN GIT AND SVN?

Git is distributed: each clone can contain full project history. SVN is centralized around a central repository.

QUESTION

WHAT HAPPENS DURING `git commit`?

Git creates a commit from the staged snapshot, records metadata and parent commit references, and updates the current branch reference.

QUESTION

`git fetch` VS `git pull`?

`fetch` downloads remote updates; `pull` fetches and integrates them into the current branch.

QUESTION

WHAT IS DETACHED HEAD?

HEAD points directly to a commit rather than a branch. To keep new commits, create a branch with `git switch -c my-branch`.

QUESTION

WHEN IS `git rebase` DANGEROUS?

Rewriting commits already shared with others can disrupt their history. Coordinate before rebasing shared work.

Save this cheat sheet for your next tech interview!

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Git

QUICK NOTES

Version Control Basics

BEGINNER TO INTERMEDIATE



Working Directory → Staging Index → Local Repository → Remote GitHub

SWIPE TO LEARN ➡

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THE THREE TREES ARCHITECTURE

"Git is not just saving files."
"It tracks snapshots of your project,
not just individual changes."



1. WORKING TREE

Files you edit directly in VS Code.

Example files:
`src/, App.js, index.js, style.css`



2. STAGING INDEX

Choose exactly which changes go into your next commit.

Command: `` `git add` ``



3. LOCAL REPOSITORY

The `` `.git` `` database stores committed project history.

Command: `` `git commit` ``

Edit files → git add → git commit

KEY COMMANDS

`git status`	`git add`	`git commit`
working-tree & staging status	stages files	persists snapshot

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